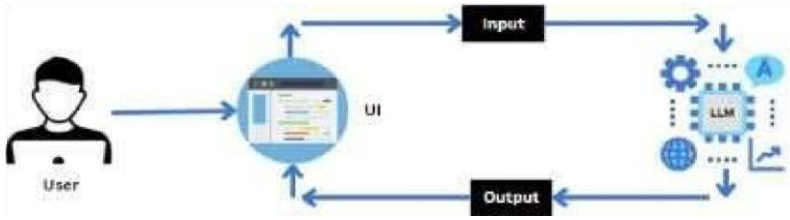
Project Design Phase
Solution Architecture

Date	11 February 2026
Team ID	LTVIP2026TMIDS90902
Project Name	Explore with AI: Custom Itineraries for Your Next Journey
Maximum Marks	2 Marks

Solution Architecture:



1. Overview

Explore with AI is a web and mobile-based platform that allows users to enter travel preferences (destination, duration, interests) and receive a personalized, AI-generated travel itinerary. The system is designed for simplicity, speed, and scalability.

2. High-Level Components

A. User Interface (Frontend)

- Technology: Streamlit (Python) or React.js for web/mobile interface
  Functionality: o Input fields for destination, trip duration, and preferences/interests o Submit button to generate itinerary o Display of AI-generated itinerary in a structured, day-wise format o Options to review, customize, or export itinerary (PDF/Text) o Display user-friendly error messages

B. API Layer / Backend

- Technology: Python (Flask/FastAPI) Functionality:
  - Receives user input from frontend
  - Validates inputs (destination, dates, preferences) o Handles requests to the AI model o Returns structured itinerary results to frontend

C. AI Engine

- Technology: Gemini Pro LLM / OpenAI GPT / other generative AI model
  - Functionality:
  - Processes input parameters (destination, duration, interests) o Generates detailed day-wise itinerary including:
    - Attractions and sightseeing
    - Local dining recommendations
    - Suggested activities and timing
    - Travel tips and optional hidden gems o Ensures structured output for easy rendering on frontend

D. Data Layer (Optional)

- Technology: Cloud database (MongoDB/PostgreSQL)
  - Functionality: o Stores predefined destination data (optional) o Stores user-generated itineraries (if history/export feature is implemented) o Logs analytics for system monitoring

E. Export & Reporting Module

- Functionality:
  - Convert itinerary to PDF or downloadable formats o Copy/export to clipboard
  - Optional integration with email or messaging for sharing

3. Key Features of Architecture

- Modular Design: Frontend, backend, AI engine, and data layer are loosely coupled.
- Scalable: Can handle more destinations, multiple users, and AI requests simultaneously.
- Extensible: Future features can include multi-destination planning, travel booking integrations, or content generation for blogs.
- Simple Workflow: Focused on input → AI processing → output, minimizing user friction.